

Recruitment failure in Florida Keys *Acropora palmata*, a threatened Caribbean coral

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Abstract Recovery of *Acropora palmata* from its currently imperiled status depends on recruitment, a process which is poorly documented in existing Caribbean coral population studies. *A. palmata* is thought to be well adapted to proliferate through the recruitment of fragments resulting from physical disturbances, such as moderate intensity hurricanes. This study monitored fifteen 150 m² fixed study plots on the upper Florida Keys fore-reef for asexual and sexual recruitment from 2004 to 2007. Between July and October 2005, 4 hurricanes passed by the Florida Keys, producing wind speeds on the reef tract of 23 to 33 m s⁻¹. Surveys following the hurricanes documented an average loss of 52% estimated live tissue area within the study plots. The percentage of “branching” colonies in the population decreased from 67% to 42% while “remnant” colonies (isolated patches of tissue on standing skeleton) increased from 11% to 27%. Although some detached branches remained as loose fragments, more than 70% of the 380 fragments observed in the study plots were dead or

rapidly losing tissue 3 weeks after Hurricane Dennis. Over the course of the study, only 27 fragments became attached to the substrate to form successful asexual recruits. Meanwhile, of the 18 new, small encrusting colonies that were observed in the study, only 2 were not attributable to asexual origin (i.e., remnant tissue from colonies or fragments previously observed) and are therefore possible sexual recruits. In summary, the 2005 hurricane season resulted in substantial loss of *A. palmata* from the upper Florida Keys fore-reef from a combination of physical removal and subsequent disease-like tissue mortality, and yielded few recruits of either sexual or asexual origin. Furthermore, the asexual and sexual fecundity of the remaining population is compromised for the near future due to the lack of branches (i.e., “asexual fecundity”) and overall loss of live tissue.

Keywords Fragment · Disturbance · Hurricane · Disease

Introduction

For species that have undergone massive population declines, enhanced successful recruitment is a necessary condition for recovery. Clonal reproduction likely enhances population persistence when environmental conditions are not conducive to successful sexual reproduction, or when adult population densities are reduced to an extent that finding mates or external fertilization is compromised (i.e., Allee effects). However, such a strategy can result in genotypically depauperate populations, which may compromise both population and ecosystem resilience in the face of continued or accelerating environmental challenges (Schmid 1994; Honnay and Bossuyt 2005; Reusch et al. 2005). Success of both sexual and asexual reproduction is likely required to regain a self-sustaining population.

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The life history strategy of acroporid corals is characterized by disturbance-induced fragmentation (Highsmith 1982; Lirman 2000; Bruckner 2002), in combination with sexual reproduction through broadcast spawning (Fadlallah 1983). Because fragmentation is linked to disturbance, clonal structure and disturbance are potentially closely linked (McFadden 1997; Coffroth and Lasker 1998; Douhovnikoff et al. 2005). As is hypothesized for species richness (Connell 1978), intermediate disturbance regimes are thought to maximize clonal propagation by optimizing both fragment production and fragment retention/survival (Coffroth and Lasker 1998; Lirman 2003).

The elkhorn coral, *Acropora palmata* (Lamark), has undergone recent widespread population declines throughout its range including the Florida reef tract, the Bahamas, and throughout the Caribbean (Bruckner 2002). In the upper Florida Keys, the shallow fore-reef was once dominated by *A. palmata* (Shinn et al. 1980; Voss 2002). By 1996 when robust coral monitoring was initiated in this region, the population had already declined by an estimated 95% from historical abundance, and percent cover was on average 7% (Miller et al. 2002b). Similar trends were observed in *A. palmata* populations throughout the Caribbean, a region profoundly affected by dramatic, albeit spatially and temporally variable, physical disturbance by hurricanes (Lugo et al. 2000; Gardner et al. 2005). The range-wide catastrophic declines from disease and hurricanes beginning in the early 1980s and continuing through the present (Bruckner 2002) resulted in its designation as a threatened species under the U.S. Endangered Species Act in 2006 (NMFS 2006).

Direct quantification of sexual and asexual recruitment success in *A. palmata* is extremely difficult. Though some studies have documented high survivorship of *A. palmata* fragments following individual storm events (Fong and Lirman 1995; Lirman and Fong 1997; Lirman 2000), baseline rates of asexual recruitment have not been quantified. Sexual recruits result from microscopic larvae settling and metamorphosing onto reef surfaces, and take several months or years to become visible in situ (D. E. Williams personal observation). Definitive *Acropora* spp. sexual recruits (i.e., settled on experimental substrates) have been observed only rarely (Van Moorsel 1988, 1989; Quinn and Kojis 2005). In other reports, successful sexual recruitment of *A. palmata* has been inferred based upon the appearance of small colonies (Highsmith 1982; Zubillaga et al. 2005; Grober-Dunsmore et al. 2006). Because most *A. palmata* populations have undergone such ubiquitous decline, many populations now comprise small, remnant tissue patches in mostly dead thickets (Bruckner 2002; Bonito and Grober-Dunsmore 2006). Much of the difficulty in distinguishing true sexual recruits results from these small remnant tissue patches, which can re-grow and develop morphology suggestive of a larval recruit (see Miller et al. 2007).

The study reported here carefully tracked individual colonies, fragments, and reef benthos over time to discriminate likely sexual recruits from other small asexually produced patches of live tissue. A plot-based study was undertaken at five fore-reef sites in the upper Florida Keys over three years, which included periods with and without physical disturbance events. By meticulously mapping the plot substrate annually, and monitoring a randomly selected subset of colonies quarterly, both the current trend in adult colony status and the effectiveness of both sexual and asexual recruitment in these remnant populations can be reported.

Materials and methods

Study sites

Study sites were chosen by targeting patches of *A. palmata* in the fore-reef spur and groove zone with intermediate densities of distinguishable, individual colonies (not thicket-type patches). Suitable plot locations (Table 1) were those with at least 10 individual colonies (Fig. 1) within a 7 m radius circle. Earlier stratified random sampling (S. Miller et al. unpublished) encompassing a range of reef habitat types found that *A. palmata* in the upper Florida Keys occurred at only 15 of 107 sampled sites (14%) making random selection of sites for this study impractical. These earlier surveys also showed *A. palmata* occurred more often and at higher abundance in fore-reef spur and groove habitats compared to other reef habitats. Colony abundance in the targeted plots ranged from 0.07 to 0.28 colonies m⁻² (Table 1) which was representative of colony abundance observed in the earlier stratified random surveys of this habitat type in the upper Florida Keys.

An iron stake was placed at the center of each plot, and 10 to 12 colonies were selected for quarterly monitoring using randomly generated coordinates relative to the center stake (bearing and distance). Colonies were numbered with small aluminum tags nailed to nearby substrate. If the center (based on top view) of the colony was outside the 7 m plot radius, it was not included. Thus, the selection of tagged colonies within plots was random, though plot placement was not. Fourteen such plots with a total of 180 tagged colonies were established among five reefs in the upper Florida Keys in May 2004 (Table 1). In September 2004, a 15th plot with 12 tagged colonies was added.

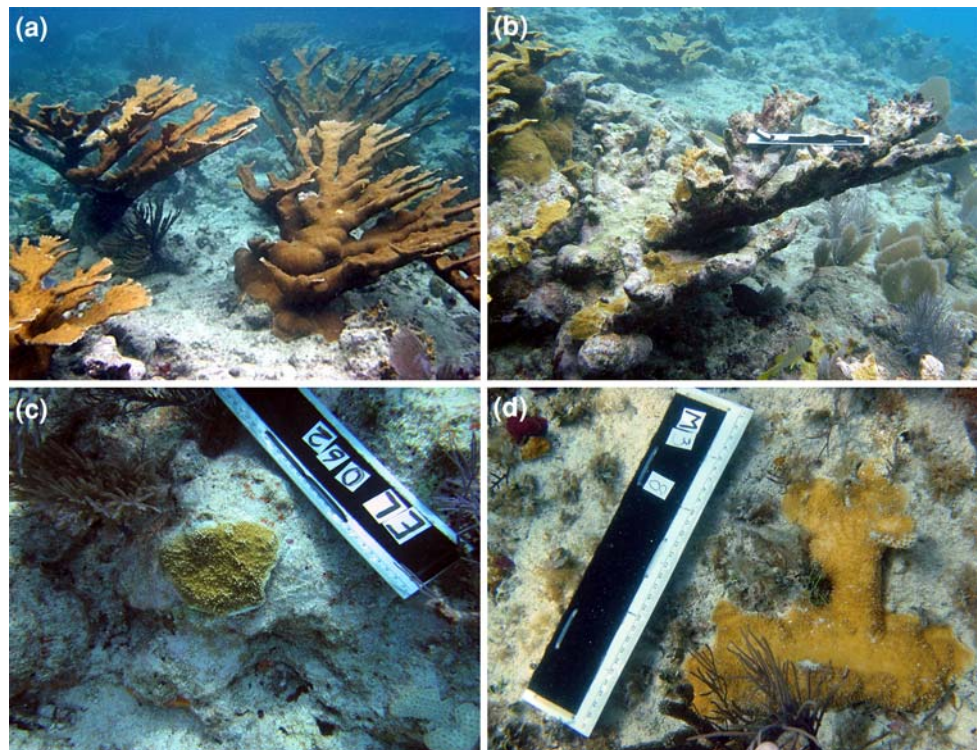
Annual plot and quarterly colony surveys

Two types of surveys were conducted in these study plots: “colony surveys” and “plot surveys.” A set of criteria for consistently delineating individual colonies was outlined and used throughout the study (Williams et al. 2006).

Table 1 Study plot attributes and live area index (LAI) at three of the plot surveys. Study plots were 7 m radius (150 m²) permanent plots established within an extant *Acropora palmata* stand and initially mapped in October 2004, and surveyed again in April 2006 and 2007. The live area index (LAI) was calculated by the average dimension of the colony squared, multiplied by the colony % live tissue cover estimated during the survey, and totaled for all colonies within the study plot

Reef	Plot	Latitude (°N)	Longitude (°W)	Depth (m)	Total no. colonies	Abundance (col m ⁻²)	Live Area Index (m ²)		
							October 2004	April 2006	April 2007
Carysfort	CF1	25°13'19.0"	80°12'38.0"	3	17	0.11	4.3	1.1	1.1
	CF2	25°13'18.4"	80°12'38.2"	3	35	0.23	9.4	6.8	7.0
	CF3	25°13'22.4"	80°12'34.4"	4	10	0.07	11.6	5.3	5.6
Elbow	EL1	25°08'33.3"	80°15'30.1"	4	42	0.28	16.0	9.0	9.0
	EL2	25°08'34.4"	80°15'29.6"	4	24	0.16	7.5	3.5	4.8
	EL3	25°08'38.2"	80°15'28.1"	5	29	0.19	6.9	2.3	2.9
	EL4	25°08'42.3"	80°15'26.4"	6	20	0.13	3.8	2.9	3.2
	EL5	25°08'42.7"	80°15'26.6"	6	13	0.09	1.7	0.7	0.7
French	FR1	25°02'02.1"	80°20'57.9"	7	28	0.19	23.1	9.3	8.0
Key Largo	KL1	25°07'25.0"	80°17'50.5"	3	17	0.11	6.7	2.3	2.3
Dry Rocks	KL2	25°07'22.4"	80°17'52.3"	2	26	0.17	3.6	0.0	0.0
	KL3	25°07'21.2"	80°17'53.7"	4	15	0.10	2.4	0.9	1.0
Molasses	ML1	25°00'34.5"	80°22'29.3"	4	11	0.07	5.5	1.8	2.0
	ML2	25°00'32.8"	80°22'29.0"	6	23	0.15	5.7	2.9	3.2
	ML3	25°00'36.5"	80°22'23.8"	4	22	0.15	9.9	7.2	7.7

Fig. 1 Examples of *Acropora palmata* colony type designations used in this study. (a) Branching colonies were those with at least two branches, at least one of which was more than half covered with live tissue; (b) Remnant colonies may have a branching structure, but less than half of the branching portion was covered with live tissue; (c) encrusting colonies did not have branches; and (d) attached fragments were colonies that originated from a loose fragment, and had fused with the substrate. Note that loose fragments (i.e., without tissue grown to adjacent substrate) were not considered “colonies”



Specifically, the following rules were applied to define an individual colony: (1) any continuous live tissue; (2) physically separated units of live tissue growing on the same underlying colony skeletal structure; (3) if live tissue was found on dead structure that was difficult to clearly differentiate into a solitary skeleton, then patches of tissue within 1 meter of each other were considered a single colony; (4) if

live tissue was subdivided by a “seam” (i.e., an obvious, raised ridge where live tissue from adjacent but physiologically discontinuous colonies meet), then the tissue on either side of the seam was considered to constitute separate colonies. Numerous photographs were taken of all colonies to ensure proper identification and consistent observations between both types of surveys.

Colony surveys were conducted quarterly beginning in April 2004 to quantify colony size, percent live tissue, colony type, and colony condition for the tagged colonies. Colony size was measured as the length (L), width (W), and height (H), where length was the maximum diameter, width was measured perpendicular to length, and height was measured from the base of the colony to the branch tips along the axis of growth. The percentage of the measured colony surface that was covered with live tissue (% live) was estimated visually in the field and confirmed by comparing photos with previous surveys.

Plot surveys were conducted approximately annually beginning in October 2004 and were repeated in June/July 2005, April 2006, and April 2007. Because the 2005 surveys were interrupted by Hurricane Dennis (9th July), some sites were not surveyed until after this event, thus the data for this survey period are not comparable and are not reported. Plot surveys consisted of searching the entire 150 m² study plot area in order to locate, photograph, map (bearing and distance from center stake), and estimate size, percent live tissue, and colony type for all attached colonies within the study plot. Colonies that were not present at the previous survey were considered new colonies, and if a colony present at one survey was not recorded at the following survey, it was considered to be “gone.” The appearance of a new “encrusting” colony in a location observed to be vacant in a previous plot survey was presumed to be a sexual recruit. Loose fragments (i.e., asexual propagules) within the plot were not mapped as “colonies” since they were not attached to the substrate.

Live tissue abundance/live area index (LAI)

A. palmata has a complex, arborescent morphology, as well as a strong tendency to display partial colony mortality and subsequent re-sheeting of the skeleton by live tissue. Although 3-D digital computer modeling techniques exist that allow for the precise calculation of coral colony live surface area, these procedures increase in cost with the morphological complexity of the colony, and are too time-consuming for routine survey of a large number of colonies (Bythell et al. 2001). Thus, field measurements of colony dimension (length, width, height) and visual estimates of percent live tissue coverage were used to compute an index of the total “abundance” of live tissue. The “Live Area Index” (LAI) was calculated as the square of the mean colony dimension, multiplied by percent live tissue $[(L + W + H)/3]^2 \times \% \text{ live}$ where L = length, W = width, and H = height. Although the units are given as m² the LAI is not intended to directly indicate surface area of the colony; instead it is an index that changes as the targeted variable (surface area) changes. LAI underestimates the actual surface area, and this discrepancy increases with increasing

complexity; thus LAI likely underestimates the tissue loss associated with a loss of colony complexity and underestimates the tissue growth associated with an increase in colony complexity. While it is not a precise measure of the area of live tissue, it is a feasible way to track the abundance of live tissue for large numbers of colonies or by whole study plots over time.

Based on annual plot surveys, the LAI for each study plot (including all colonies within the plot) was calculated (Table 1). For increased temporal resolution, the LAI for each tagged colony was pooled by plot for each quarterly colony survey and compared among reefs and over time using a repeated measures ANOVA. Significant factors were compared using Tukey’s HSD post hoc comparison.

Colony type

Colony types were classified as “branching,” “remnant,” “encrusting,” or “attached fragment” (Fig. 1). Branching colonies (Fig. 1a) were those with at least two branches, at least one of which was more than half-covered with live tissue; remnant colonies (Fig. 1b) may have a branching structure, but less than half of the branching portion was covered with live tissue. Encrusting colonies (Fig. 1c) did not have branches, and attached fragments (Fig. 1d) were colonies that originated from a loose fragment, and had fused with the substrate.

To compare changes in the distribution of colony types in the overall population, the numbers of branching, remnant, and “other” colony types (encrusting plus attached fragments) were pooled over all plots and compared with separate McNemar’s tests for each time interval (i.e., between October 2004 and April 2006, and between April 2006 and April 2007). A Bonferroni adjusted *p*-value of <0.025 was used to judge significance for these two tests.

Fragment survivorship

In a supplemental survey conducted 4–5 days after Hurricane Dennis passed (9th July 2005), the loose fragments in each study plot were counted. During the regular quarterly survey conducted 22–30 July 2005, the loose fragments were re-counted and classified as either healthy or dying (recent loss of $\geq 1/3$ of live tissue).

Results

Live tissue abundance

The initial total LAI of the fifteen 150 m² study plots ranged from 1.7 to 23.1 m², and subsequently decreased steadily in all plots (median = 66%, inter-quartile

range = 22%, $n = 15$) from October 2004 to April 2006 (Table 1). Between April 2006 and April 2007, the overall decline in LAI slowed with some variability between plots. The decrease in LAI over the entire study ranged from 15% to 99% among plots (Table 1).

The LAI of the tagged colonies surveyed quarterly showed significant temporal variation (ANOVA $p < 0.0001$, $df = 9$) but no significant differences among reefs. Significant declines in LAI occurred between May and August of 2005 (Tukey HSD $p = 0.0002$) following Hurricane Dennis, and between August 2005 and January 2006 (Tukey HSD $p = 0.0306$) following Hurricanes Katrina, Rita, and Wilma (Fig. 2).

Colony abundance and turnover

The number of colonies present at the initial plot survey (October 2004) totaled 331 in the 15 study plots. By April 2006 and April 2007 plot surveys, this count decreased to 274 and 262, respectively. While the net loss was 69 colonies, the intermittent plot surveys documented more dynamic colony turnover (Fig. 3). Only 196 of the 331 originally mapped colonies were intact and at least partially alive by the April 2007 plot survey. Over this time period, 133 colonies experienced complete mortality, or were physically removed from the plot. A total of 63 “new” colonies were counted within the plots between October 2004 and April 2007. Included in these 63 “new” colonies were 34 “colony vestiges.” These were colonies that were present in the same location throughout the study but, because of fragmentation, they could no longer reasonably be considered part of the original colony. Thus, these “new” colonies did

not represent novel recruits to the population, but would likely be considered a separate individual colony in an initial survey according to the colony definition rules (see “Materials and methods”). Additionally there were 27 new attached fragments, and only 2 new colonies that were classified as sexual recruits. While it is not certain that they originated from planulae, they were the only new colonies that were not directly observed to originate by asexual means.

Fragment survivorship

Following Hurricane Dennis the number of loose fragments in each plot ranged from 0 to 113 and totaled 380. Within two to three weeks of the storm an average of 71% of those fragments were dying (recent loss of one-third of their live tissue) or recently dead (white freshly exposed skeleton) in situ. Although some were clearly abraded, many of the fragments that were dying showed no signs of abrasion; all corallites and the remaining tissue appeared undamaged.

Colony type

There was a significant change (McNemar, $p < 0.001$, $df = 3$) in the distribution of colony types for each time interval examined (October 2004–April 2006 and April 2006–April 2007). The number of colonies that were classified as “branching” decreased from 222 in the first plot survey in October 2004 to 111 in April 2006, following the hurricanes. By April 2007 this increased to 144 (Fig. 4). The loss of branches did not result in a significant increase in attached fragment type colonies, which remained relatively constant at approximately 7–10% of the population (Fig. 4). Rather, the “branching” colonies became “remnant” colonies which increased from 36 at the first survey to 73 in April 2006 and then decreased to 30 by April 2007 as some of these remnant colonies began to branch.

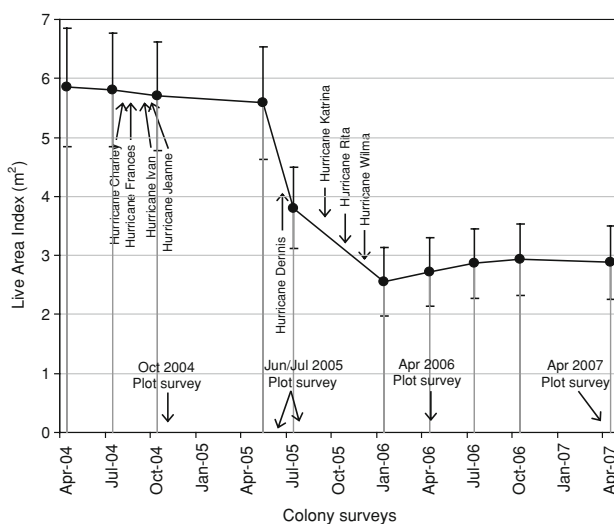


Fig. 2 Live Area Index (LAI) for the tagged *Acropora palmata* colonies summed by plot and averaged for all plots ($n = 15$, mean ± 1 SE). The timing of each hurricane and plot survey is shown by arrows

Discussion

In the 3 years duration of this study, there was a substantial loss of *A. palmata* within all monitored plots in the upper Florida Keys. The loss was apparent in both the amount of live tissue, which decreased by more than half, and in the total number of live colonies, which decreased by over one-third. Based on the tagged colony surveys, the storms of the 2005 hurricane season (July–November) detached 32 (out of 180) whole colonies and either removed them from the survey plots or shattered them into numerous fragments observed in the study plots after the storms. However, 20 tagged colonies suffered complete tissue mortality while remaining at least partly in place following the storms.

Fig. 3 Total number of *Acropora palmata* colonies in April 2007 in each plot. Colonies that died or were physically removed between October 2004 and April 2007 are grouped as “gone” and plotted as negative values. “Surviving Colonies” are those that were alive for the initial survey, and were still recognized as living colonies in April 2007. “Colony Vestiges” represent living *A. palmata* tissue that was initially considered part of another mapped colony, but due to breakage or mortality could no longer be defined as a single colony (see Fig. 1). Thus they do not represent any new living material. In contrast, “Likely Sexual Recruits” and “New Attached Fragments” represent recruitment to the population

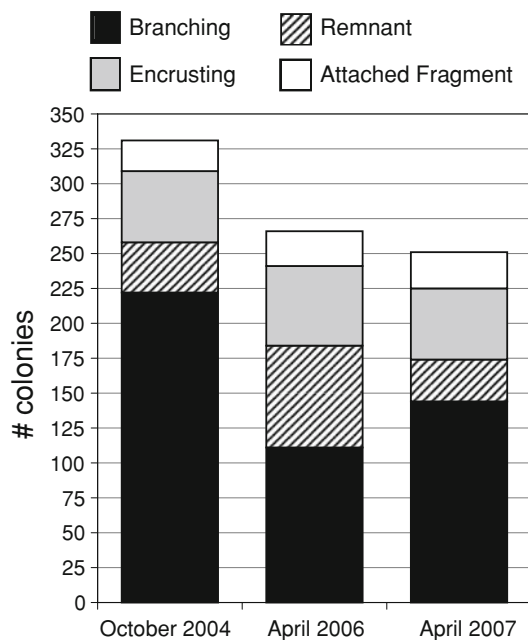
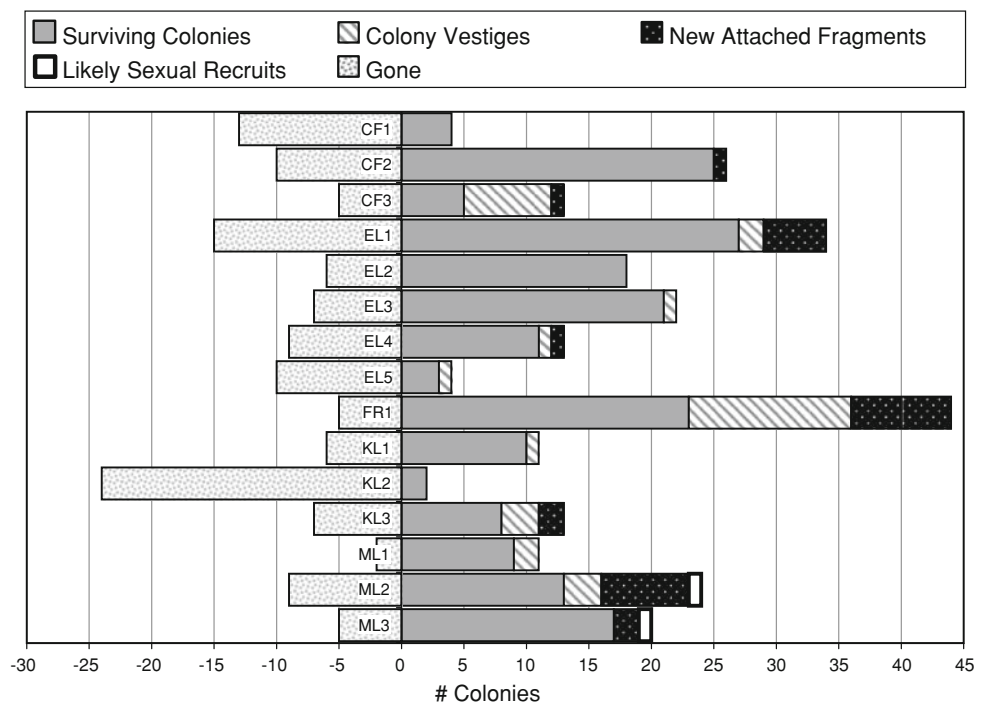


Fig. 4 Total number of *Acropora palmata* colonies (pooled across all plots) classified by type at each plot survey

Within four days of Hurricane Dennis, tissue loss was observed along the sides of the branches as though the tissue had been scoured off by storm-suspended particulates (Fig. 5), yet progressed over the following weeks in a manner more consistent with the “white-syndrome” of unknown etiology observed commonly in the upper Florida Keys since 2003 (Williams and Miller 2005, 2006; Williams et al. 2006).

Fragmentation is a major component of *A. palmata*'s life history (Highsmith et al. 1980; Fong and Lirman 1995), and was an expected and potentially beneficial (by creating potential asexual propagules) result of the storms. The immediate storm-related declines and shifts in colony type observed in this study are consistent with model predictions of Lirman (2003), while the subsequent recruitment and recovery observed was much poorer than predicted. Based on Lirman's (2003) population model, *A. palmata* was predicted to decline in abundance and shift from colonies to loose fragments and crusts from immediate storm impacts. When modeled storm frequency was less than every 5 years, the modeled population displayed a positive abundance trend (Lirman 2003) via recruitment of fragments. Although this model suggests that declines immediately following a hurricane do not predict longer-term population trajectory, this scenario was based on previous (1993–1997) upper Florida Keys field data suggesting 15% of storm-generated fragments recruit to colonies (Lirman 2003). The fragment recruitment observed in the current study was <5% (18 of at least 380) following the 2005 hurricanes.

Sexual recruitment rates of *A. palmata* have always been assumed to be low; in spite of the large number of settlement plate studies conducted in the Caribbean, only a few have found *Acropora* spp. recruits (Rogers et al. 1984; Van Moorsel 1988; Kojis and Quinn 2001; Quinn and Kojis 2005). Other studies have reported sexual recruitment in *Acropora* spp. based on the presence of small encrusting colonies presumed to be sexual recruits (Jordan-Dahlgren 1992; Zubillaga et al. 2005; Grober-Dunsmore et al. 2006).

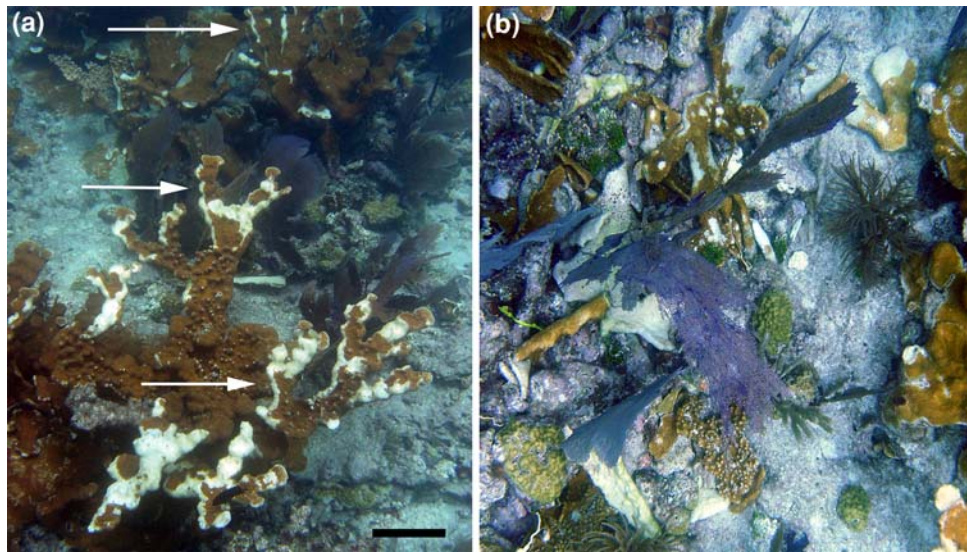


Fig. 5 Recent tissue loss observed on *Acropora palmata* colonies and fragments following Hurricane Dennis. **(a)** The tissue loss was particularly acute on the southeast sides of the branches of standing colonies (at arrows, consistent with the prevailing wind direction of the storm). However, the predominance of tissue loss was reminiscent of the white syndrome that is commonly observed affecting *Acropora* spp. in the

Florida Keys independently of storm events. **(b)** Similar tissue loss affected fragments following the storm. Several of the white fragments in this photograph were at least partially alive 4 days following the storm. The rapid mortality of fragments has also been observed during this study, independent of storm events. Black scale bar is approximately 30 cm in both frames

However, the findings of the present study indicate that this practice can grossly overestimate recruitment of larval origin (see also Miller et al. 2007). Small (3–4 cm) encrusting colonies often resulted from asexual processes (e.g., remaining basal crust from a hurricane-dislodged colony), but the frequency of surveys in the current study allowed them to be recognized as asexual recruits (Fig. 6). Out of the 18 new encrusting colonies that appeared, only two

could not be attributed to asexual origin based on observations, and these were cautiously interpreted to be sexual recruits. Thus in the 2,250 m² area of *Acropora* spp. habitat that was monitored, three spawning events (D. E. Williams personal observation) have produced only 2 potential sexual recruits that have survived to a detectable size in 3 years. Growth rates of recruits could produce a detectable encrusting colony as early as 8 months after settlement

Fig. 6 Illustration showing asexual origin of small encrusting *Acropora palmata* colonies resembling sexual recruits; fragmentation of an attached fragment (colony) leaves only the basal attachment crusts remaining. Three small individual branching colonies result 1 year later



(M. W. Miller personal observation), but given the abundance of cryptic habitats, detection may not occur until the recruit is much older. Thus, larval recruitment rates could not be determined rigorously due to unknown post-settlement mortality before they could be detected. However, the rate of larval recruitment and survival to ≥ 3 cm diameter is clearly rare in the Florida Keys, and substantially rarer than the appearance of small (<10 cm) encrusting colonies on the reef.

In addition to the low observed recruitment, other aspects of the present Florida Keys population suggest that its recruitment potential is compromised, at least in the near-term. First, the dramatic decrease in the abundance of live tissue and the decreased colony size will necessarily decrease the fecundity and spawning potential of the remaining colonies (Soong and Lang 1992; Lirman 2000). In terms of asexual reproduction, the smaller remaining colonies have fewer and smaller branches (Fig. 4) available to produce fragments; in a sense, this can be considered reduced asexual fecundity. Second, the population is comprised more of “remnant colonies” than at the start of the study, meaning that even if they have a branched structure, fragments (i.e., asexual “propagules”) that are generated are more likely to have only partial tissue cover, thus are less “viable” due to their reduced probability of fusing with the substrate. Third, the loss of whole colonies increases the likelihood of Allee effects limiting fertilization success of gametes for this outcrossing, externally fertilizing species (Baums et al. 2005). Finally reduced colony density and structural complexity can result in decreased fragment retention (Lirman and Fong 1997; Baums et al. 2003) as fragments are more likely to be swept into unsuitable habitat (D. E. Williams personal observation) thus further reducing asexual recruitment.

With substantial and ongoing decreases in the adult population, *Acropora* spp. recovery or even persistence has become increasingly dependant on successful recruitment. The observations reported here document a recent failure in asexual as well as larval recruitment in the upper Florida Keys. Given that fragmentation is the primary reproductive strategy for *A. palmata* (Bak and Criens 1982; Bak 1983; Lirman 2000; Miller et al. 2007), the relative failure of this process coupled with the confirmed rarity of sexual recruitment have alarming consequences for population recovery. Similar recruitment failure coinciding with increasing adult mortality among foliaceous Caribbean coral species less prone to fragmentation (e.g., Hughes and Tanner 2000) resulted in a pattern that resembled an “extinction vortex,” a term used to describe accelerating feedbacks of population decline (Gilpin and Soulé 1986; Fagan and Holmes 2006). Data in the current study document similar coincidence of failure of adult persistence with reduction and/or failure of both sexual and asexual reproductive modes for *A. palmata* in the upper Florida Keys.

In light of the drastically reduced “baseline” of Florida Keys’ *A. palmata* populations at the start of this study (Miller et al. 2002a), the prognosis is bleak for Florida Keys populations and underscores the crucial need for the rapid development and implementation of effective recovery measures to reduce manageable sources of adult mortality and enhance reproductive success.

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